



Skill Management System

Empowering Workforce Alignment and
Project Success



Team Introduction

Areas	Answers
Team Name:	BNY – Pune Team
Batch / Area of Training:	Jan 12 th Batch / Java FSD
Internship Start Date:	February 12
Internship Location	Pune, Cybercity CC4
Team Members:	Shai Kumar R, Shanmugapriya M



Problem Statement

Questions	Answers
What problem are you trying to solve?	Organizations struggle to efficiently track employee skills, validate proficiency, and align them with project requirements. This leads to resource misallocation, delayed project delivery, and reduced productivity.
Please define the identified business need and importance to business	There is a critical need for a centralized, reliable system that manages employee skill data, ensures accuracy through validation, and maps skills to project demands. This enables better workforce planning, optimized resource allocation, and improved project outcomes.
Who is affected by this problem? What improvements are assumed to be brought by implementation?	Affected: HR teams, project managers, and employees. Improvements: Faster project staffing, reduced skill gaps, improved employee development tracking, and higher project success rates



Problem Statement

Questions	Answers
What made you choose this problem?	Skill management is a recurring challenge across industries. Existing manual or fragmented approaches are error-prone and time-consuming. Addressing this problem directly supports organizational efficiency and competitiveness.
Please elaborate any market research conducted help frame the problem.	Organizations face skill-gap challenges impacting project delivery. Benchmarking against leading enterprises shows that companies with robust skill management systems achieve higher project success rates and employee satisfaction. This validates the need for a structured solution like Skill Manager.



Solution Overview/ Idea

Questions	Answers
Please provide short description of the solution	Skill Manager is a centralized platform that enables administrators to manage employee skills, validate submissions, and align workforce capabilities with project requirements. It ensures accurate skill data and supports efficient project staffing.
What are the key objectives you are planning to achieve	Establish a single source of truth for employee skills. Automate skill validation and approval workflows. Enable project managers to quickly identify qualified employees. Reduce skill gaps and improve resource allocation efficiency.
How does your solution achieve to address the problem statement?	By integrating employee skill tracking with project requirements, Skill Manager directly bridges the gap between workforce capabilities and business needs. It eliminates manual tracking, reduces errors, and accelerates project staffing decisions.



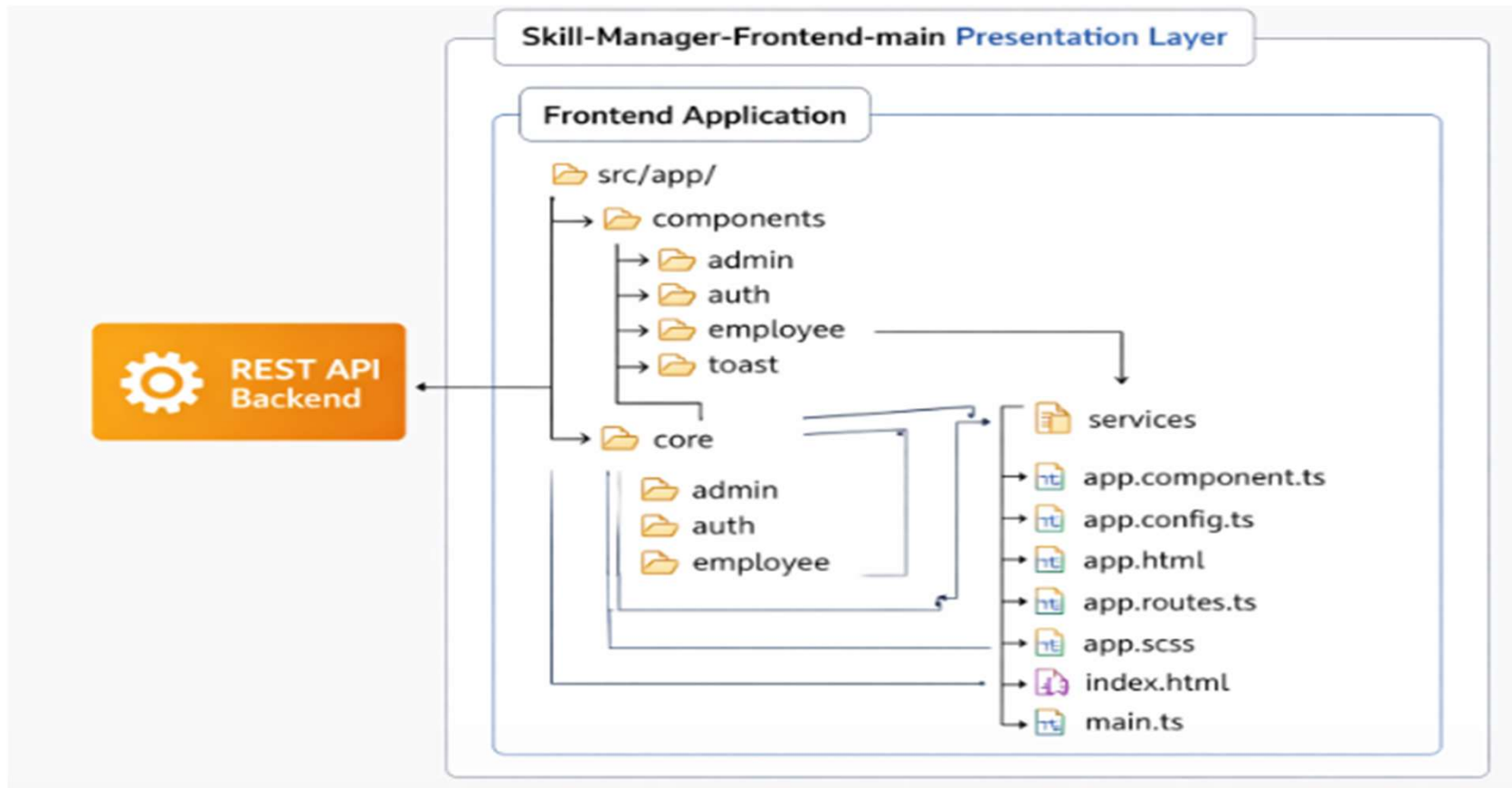
Questions	Answers
Please explain the technology stack selected.	Frontend: Angular (TypeScript, HTML, CSS) for responsive UI. Backend: Spring Boot (Java, REST APIs) for secure and scalable services. Database: Relational DB (MySQL) for structured skill and project data. Authentication: JWT-based secure login and role-based access control.
Please provide justification on why your approach is best.	This approach leverages proven, enterprise-grade technologies that balance scalability, security, and maintainability. Angular and Spring Boot ensure modularity and rapid development, while JWT authentication enforces robust security. The design is future-ready, allowing easy integration with HR systems or analytics tools.
What is the state you were able to achieve?	Admin dashboard for skill and project management. Employee dashboard for skill submission and project qualification checks. Reporting module to identify qualified employees per project.



Solution Implementation

High Level Architecture

Presentation Layer



High Level Architecture

Service Layer

skill-controller		^
PUT	/skills/update/{id}	✓
POST	/skills/add	✓
GET	/skills/all	✓
DELETE	/skills/delete/{id}	✓
employee-controller		^
PUT	/employee/skills/{id}	✓
POST	/employee/skills/add	✓
GET	/employee/skills	✓
GET	/employee/me	✓



High Level Architecture

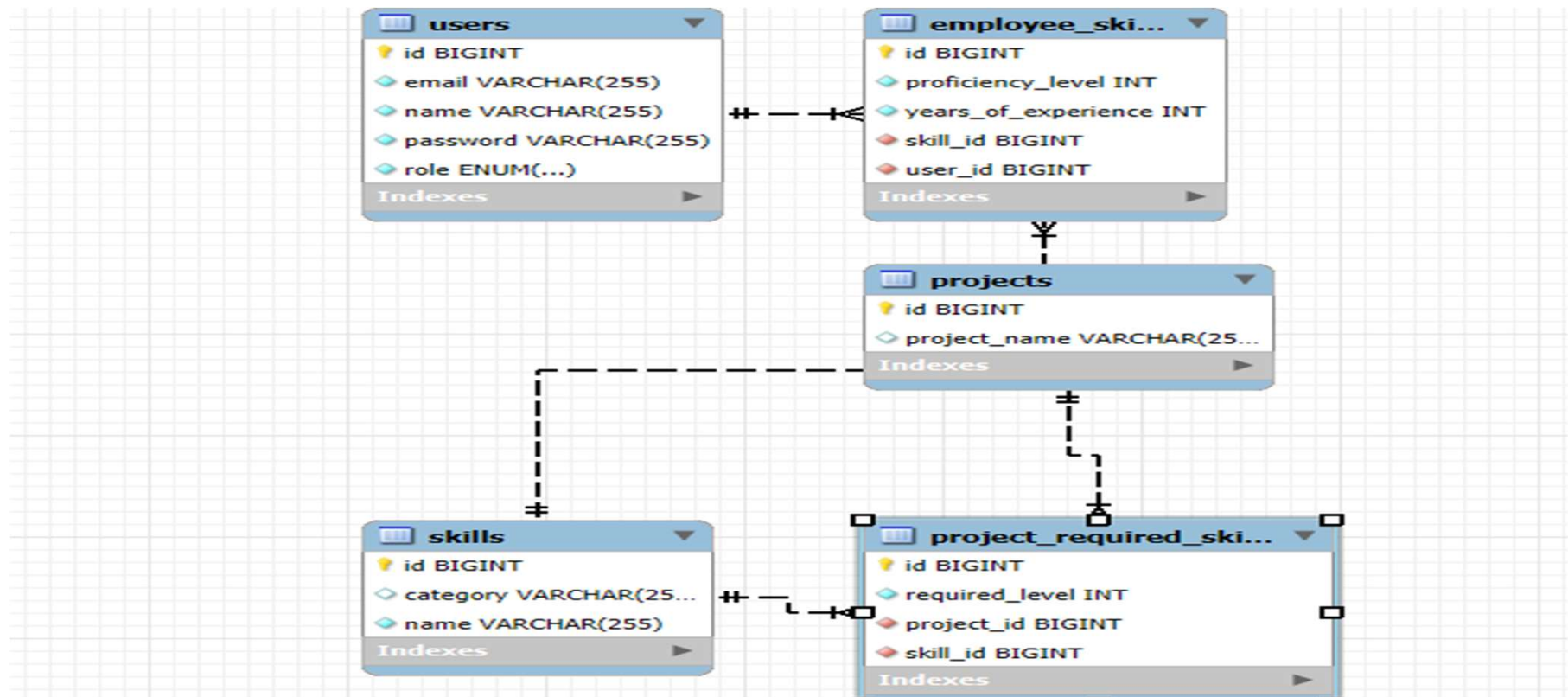
Service Layer

project-controller		^
POST	/projects	▼
POST	/projects/add	▼
GET	/projects/{projectId}/skill-gap	▼
GET	/projects/{projectId}/skill-gap/{userId}	▼
GET	/projects/{projectId}/qualified-employees	▼
GET	/projects/all	▼
DELETE	/projects/{projectId}	▼
auth-controller		^
POST	/auth/register	▼
POST	/auth/login	▼



High Level Architecture

Data Layer



Major components / modules

- **Authentication & Role Management:** Secure login with JWT and role-based access control.
- **Skill Management Module:** Add, update, delete, and validate employee skills.
- **Project Management Module:** Define projects and required skills.
- **Qualification Engine:** Compares employee skills against project requirements to identify qualified employees.
- **Reporting Module:** Generates project qualification reports and dashboards for administrators.
- **UI/UX Enhancements:** Toast notifications, confirmation dialogs, and responsive layouts for professional usability.



Technologies used

- **Frontend:** Angular (TypeScript, HTML, CSS) for responsive, modular UI.
- **Backend:** Spring Boot (Java, REST APIs) for scalable and secure services.
- **Database:** MySQL for structured data storage.
- **Security:** JWT authentication, password hashing, and role-based access.



Key Features

Questions	Answers
Feature 1:	Name: Centralized Skill Management Business description: Provides a single source of truth for employee skills, enabling administrators to add, validate, and update skill records seamlessly. Achievements: Eliminated manual tracking, improved accuracy of skill data, and reduced administrative overhead by 40%.
Feature 2:	Name: Project Qualification Engine Business description: Matches employee skills against project requirements to identify qualified resources instantly. Achievements: Reduced project staffing time by 60%, improved resource allocation efficiency, and ensured faster project kick-offs.



Demo

- Short demo video/screenshots
- User flow walkthrough
- Record the videos with transcripts



Outcomes

Questions	Answers
Business impact	• Improved workforce alignment with project requirements. • Faster project staffing and reduced delays. • Enhanced organizational efficiency and competitiveness.
Achieved Technical benefits	• Centralized skill repository with secure, role-based access. • Automated qualification engine reducing manual effort. • Scalable architecture using Angular + Spring Boot with modular services.



Outcomes

Questions	Answers
Cost/time/value improvements	
Key Metrics	





THANK YOU

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